A piston and cylinder machine contains a fluid system which passes through a complete cycle of four processes. During a cycle, the sum of all heat transfers is − 170 kJ. The system completes 100 cycles per min. Complete the following table showing the method for each item, and compute the net rate of work output in kW.

Process	Q *kJ min*	W *kJ min*	ΔE *kJ min*
a–b	0	2,170	—
b–c	21,000	0	—
c–d	–2,100	—	–36,600
d–a	—	—	—

Solution Process *a–b*:

$$Q = \Delta E + W$$
$$0 = \Delta E + 2170$$
$$\therefore \quad \Delta E = -2170 \text{ kJ/min}$$

Process *b–c*:

$$Q = \Delta E + W$$
$$21{,}000 = \Delta E + 0$$
$$\therefore \quad \Delta E = 21{,}000 \text{ kJ/min}$$

Process *c–d*:

$$Q = \Delta E + W$$
$$-2100 = -36{,}600 + W$$
$$\therefore \quad W = 34{,}500 \text{ kJ/min}$$

Process *d–a*:

$$\sum_{\text{cycle}} Q = -170 \text{ kJ}$$

The system completes 100 cycles/min.

$$\because \quad Q_{ab} + Q_{bc} + Q_{cd} + Q_{da} = -17{,}000 \text{ kJ/min}$$
$$0 + 21{,}000 - 2{,}100 + Q_{da} = -17{,}000$$
$$\therefore \quad Q_{da} = -35{,}900 \text{ kJ/min}$$

Now $\oint dE = 0$, since cyclic integral of any property is zero.

$$\therefore \quad \Delta E_{a-b} + \Delta E_{b-c} + \Delta E_{c-d} + \Delta E_{d-a} = 0$$
$$-2{,}170 + 21{,}000 - 36{,}600 + \Delta E_{d-a} = 0$$
$$\therefore \quad \Delta E_{d-a} = 17{,}770 \text{ kJ/min}$$
$$\therefore \quad W_{d-a} = Q_{d-a} - \Delta E_{d-a}$$
$$= -35{,}900 - 17{,}770 = -53{,}670 \text{ kJ/min}$$

The table becomes

Process	Q (kJ/min)	W (kJ/min)	ΔE (kJ/min)
a–b	0	2,170	– 2,170
b–c	21,000	0	21,000
c–d	–2,100	34,500	– 36,600
d–a	–35,900	– 53,670	17,770

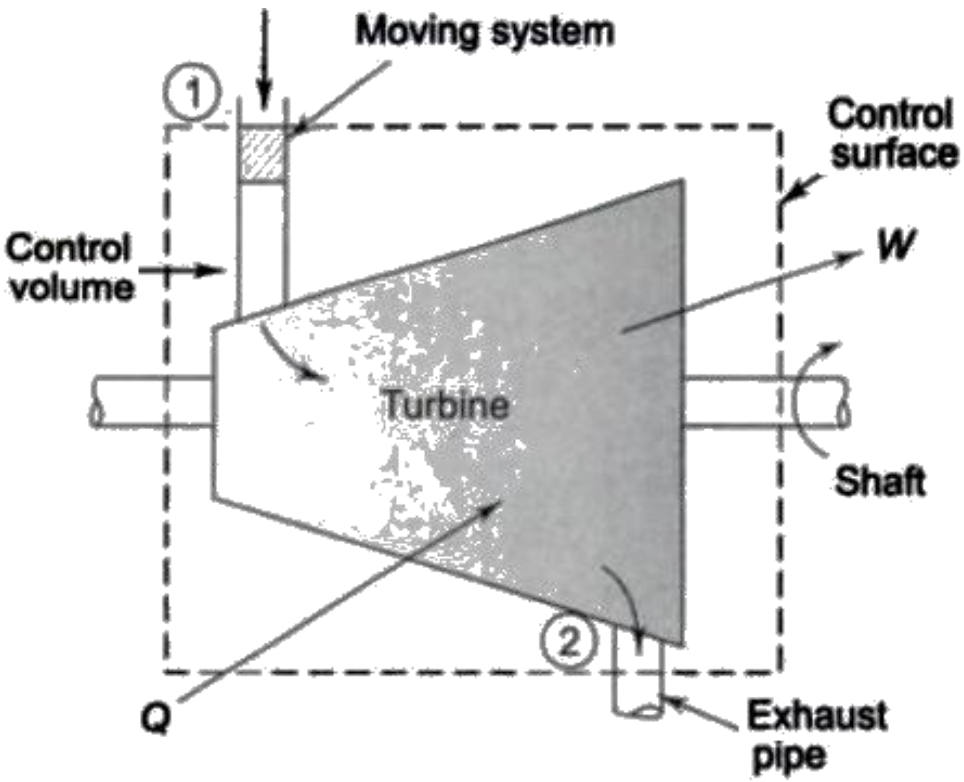
Since

Rate of work output $$\sum_{\text{cycle}} Q = \sum_{\text{cycle}} W$$

$$= -17{,}000 \text{ kJ/min} = -283.3 \text{ kW}$$

First Law of Thermodynamics applied to Flow processes

Control Volume & Control Surface



A ***control volume*** is a certain fixed region on which attention is focused and through which the moving substance flows.

A ***control surface*** is the surface on the control volume.

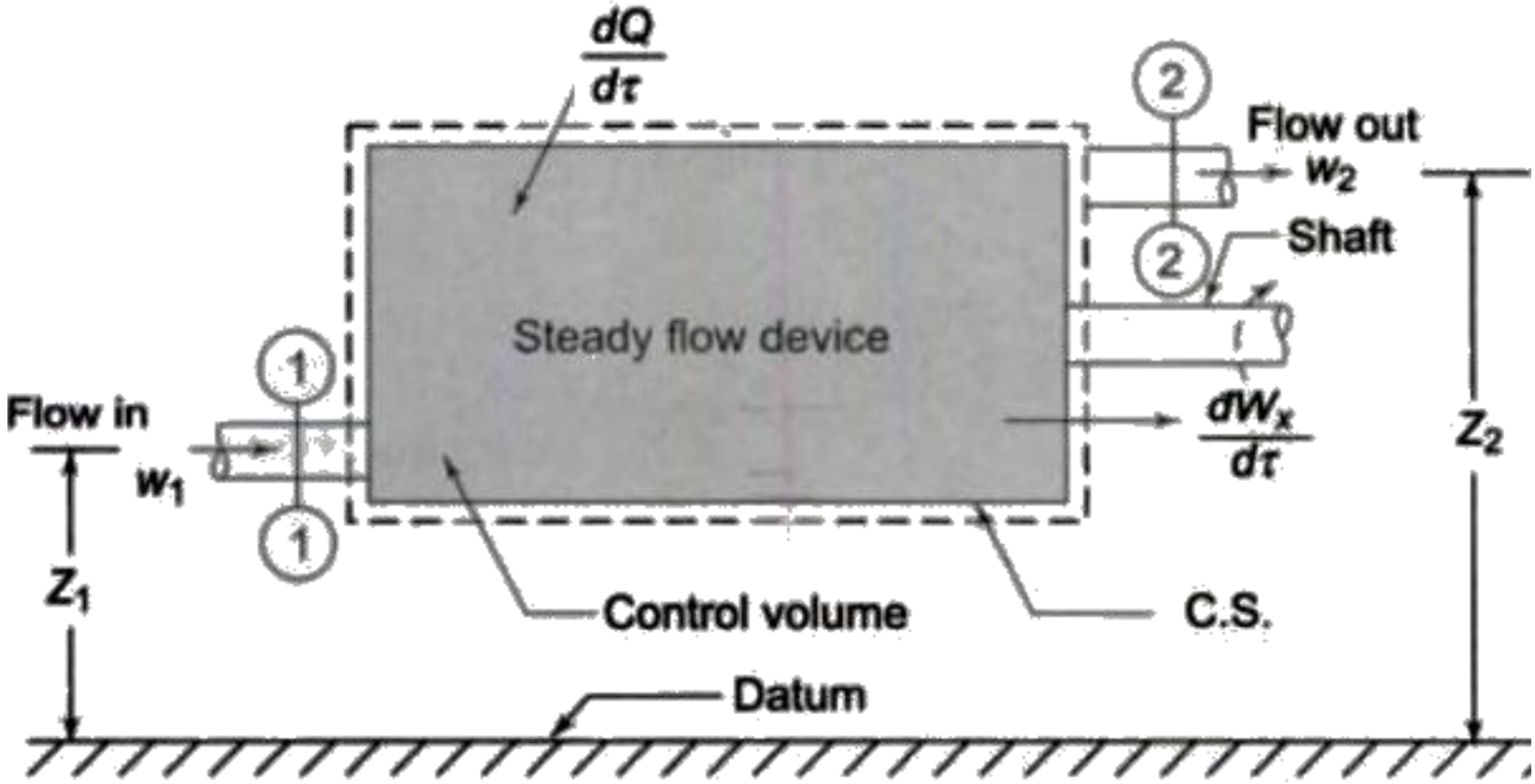
Steady flow process

- The flow is said to be ***steady*** if the rate at which the fluid flows through a machine or piece of apparatus and the rate at which the energy changes are constant.
- At the steady state of a system, any thermodynamic property will have a fixed value at a particular location, and will not alter with time. These thermodynamic property depends only on space coordinates but not on time.

Assumptions made in the analysis of steady flow process of a system

1) The mass flow through the system remains constant.
2) Fluid is uniform in composition.
3) The only interaction between the system and its surroundings are work and heat.
4) The state of fluid at any point remains constant with time.
5) In the analysis, only potential, kinetic and flow energies are considered.

Simple steady flow process of a system



The two governing equations are:

1) ***Mass balance equation***, and
2) ***Energy balance equation***

Notations

A	-	Cross-sectional area, m^2
$\dot{m}$	-	Mass flow rate, *kg/sec*
p	-	Absolute pressure, N/m^2
v	-	Specific volume, m^3/kg
u	-	Specific internal energy, *J/kg*
C	-	Velocity, *m/sec*
Z	-	Elevation above an arbitrary datum, *metres*
$\dot{q} = \frac{dQ}{d\tau}$	-	Net rate of heat transfer through the control surface, *J/sec*
$\dot{W} = \frac{dW}{d\tau}$	-	Net rate of work transfer through the control surface, *J/sec*
τ	-	Time, *sec*
$subscript - 1$	-	Inlet conditions
$subscript - 2$	-	Outlet conditions

Mass balance equation

By the conservation of mass, if there is no accumulation of mass within the control volume, the mass flow rate entering the system must be equal to the mass flow rate leaving the system. i.e.

mass flow rate entering = mass flow rate leaving

$$\dot{m}_1 = \dot{m}_2$$

$$\boxed{\frac{A_1 C_1}{v_1} = \frac{A_2 C_2}{v_2}}$$

This equation is known as ***Mass balance equation***

(or) ***Continuity equation***

Work transfers in steady flow processes

- In the flow process, the work transfer may be an ***external work*** (or) ***a flow work.***
- The external work refers to all the work transfer across the control surface other than that due to normal fluid forces.
- The only external work transfers considered in engineering thermodynamics are ***shear work (shaft*** or ***stirring)*** and ***electrical work.***

Energy balance equation

Since there is no accumulation of energy, by the conservation of energy, the total rate of flow of all energy streams entering the control volume must be equal to the total rate of flow of all energy streams leaving the control volume. i.e.

Total energy flow rate entering = Total energy flow rate leaving

$$Q + U_1 + p_1V_1 + + \frac{1}{2}mC_1^{\ 2} + mgZ_1 = W + U_2 + p_2V_2 + + \frac{1}{2}mC_2^{\ 2} + mgZ_2$$

(or) $$Q + H_1 + \frac{1}{2}mC_1^{\ 2} + mgZ_1 = W + H_2 + \frac{1}{2}mC_2^{\ 2} + mgZ_2$$

Considering unit mass of the substance,

$$\boxed{\frac{Q}{m} + h_1 + \frac{C_1^{\ 2}}{2} + gZ_1 = \frac{W}{m} + h_2 + \frac{C_2^{\ 2}}{2} + gZ_2}$$

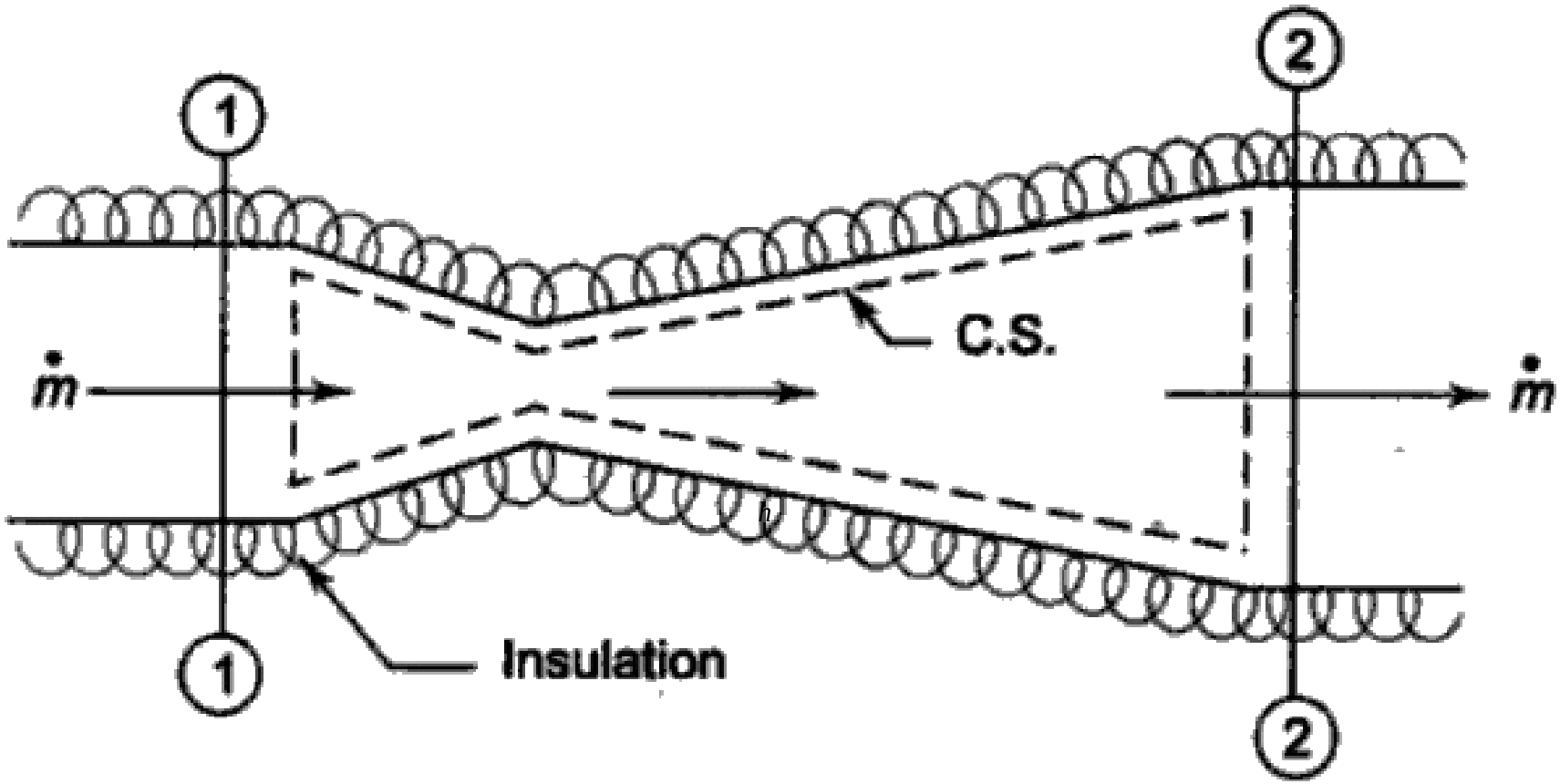
This equation is known as ***Energy balance equation***

(or) ***Steady Flow Energy Equation (SFEE)***

Applications of First law applied to flow processes

1) Nozzle
2) Diffuser
3) Compressor
4) Turbine
5) Throttling device
6) Boiler
7) Evaporator
8) Condenser
9) Water turbine
10) Water pump

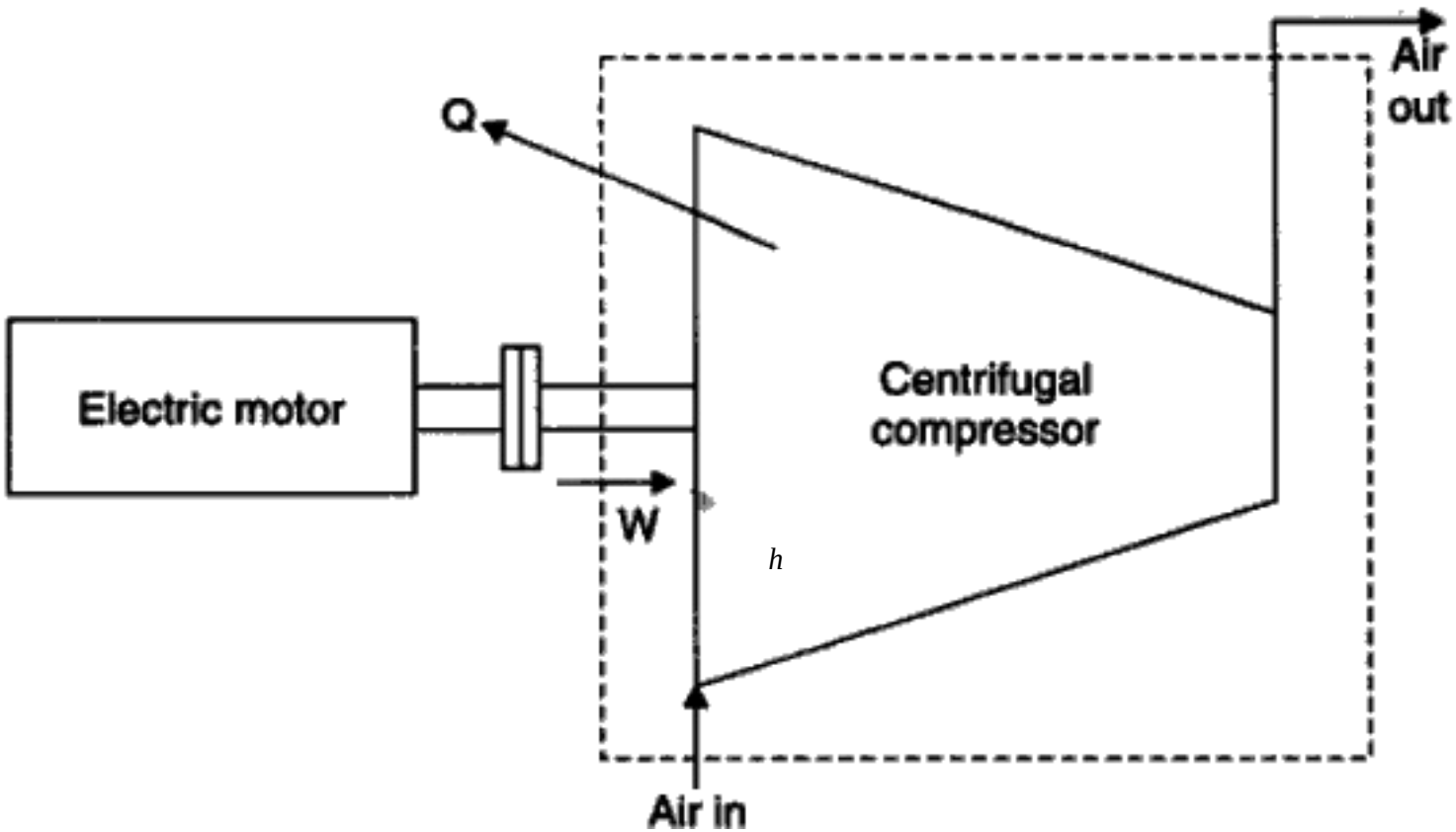
Nozzle / Diffuser



Nozzle Diffuser

$$h_1 + \frac{C_1^{\,2}}{2} = h_2 + \frac{C_2^{\,2}}{2}$$

Compressor

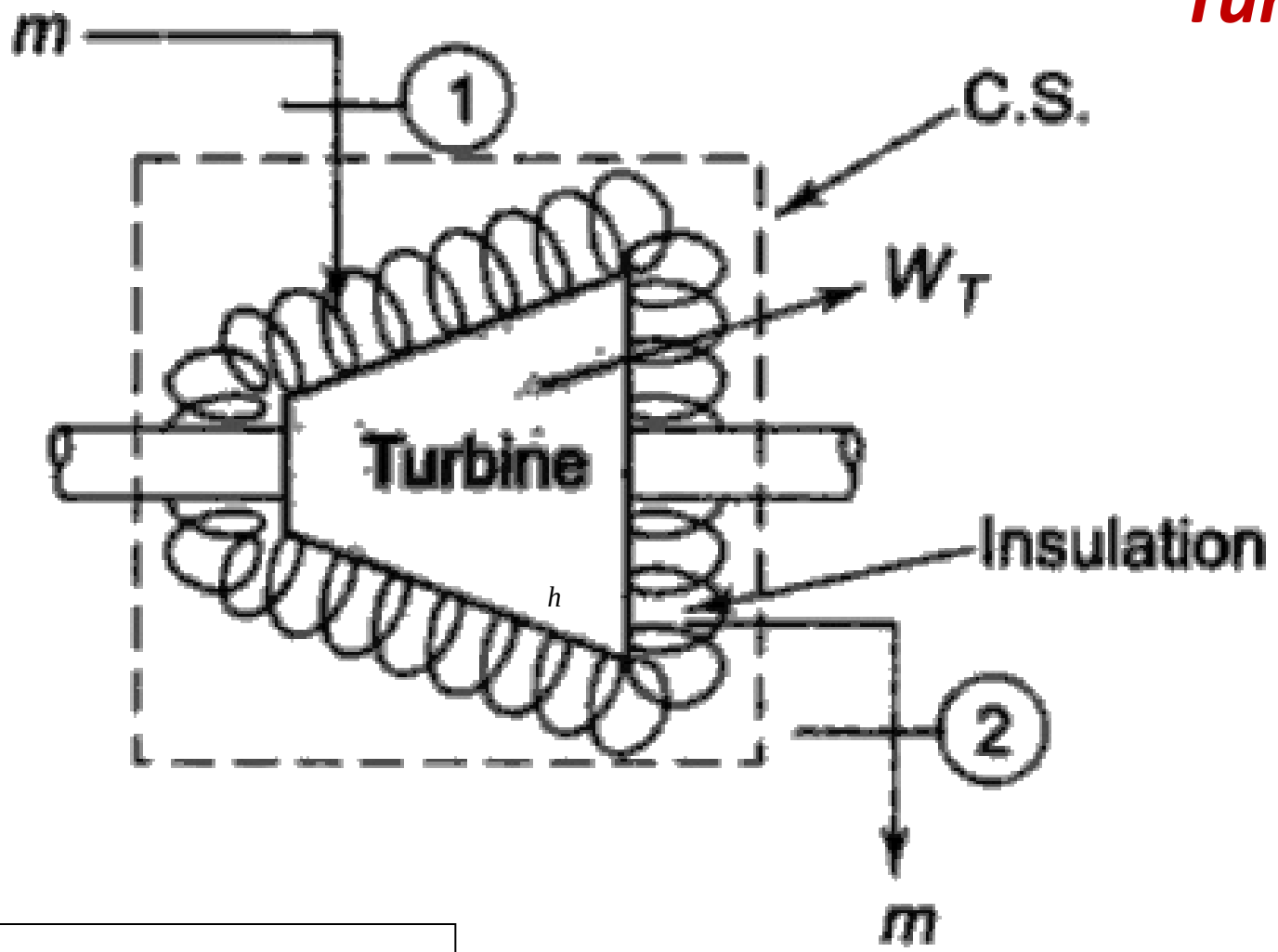


$$h_1 + \frac{W}{\dot{m}} = h_2$$

(or)

$$W = \dot{m}(h_2 - h_1)$$

Turbine

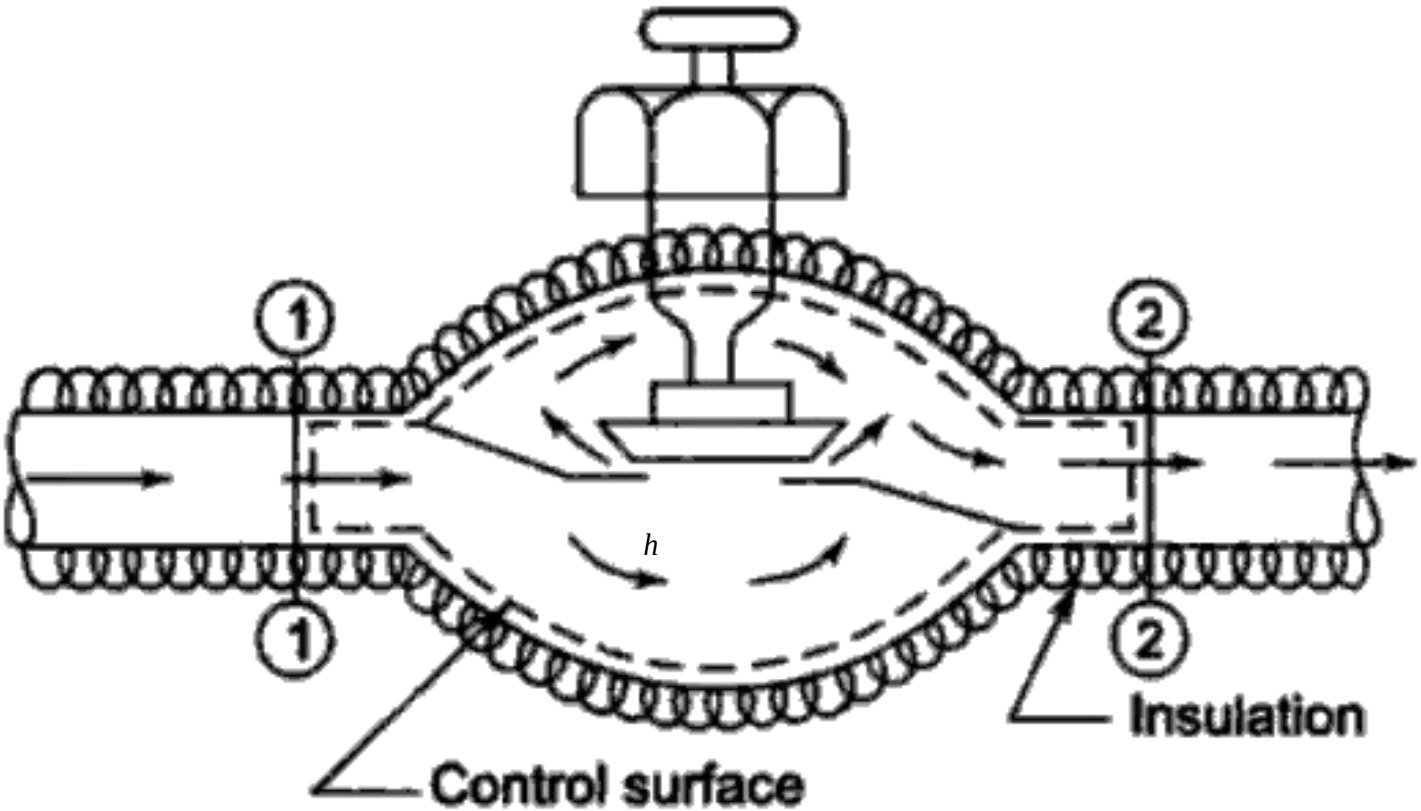


$$h_1 = h_2 + \frac{W}{\dot{m}}$$

(or)

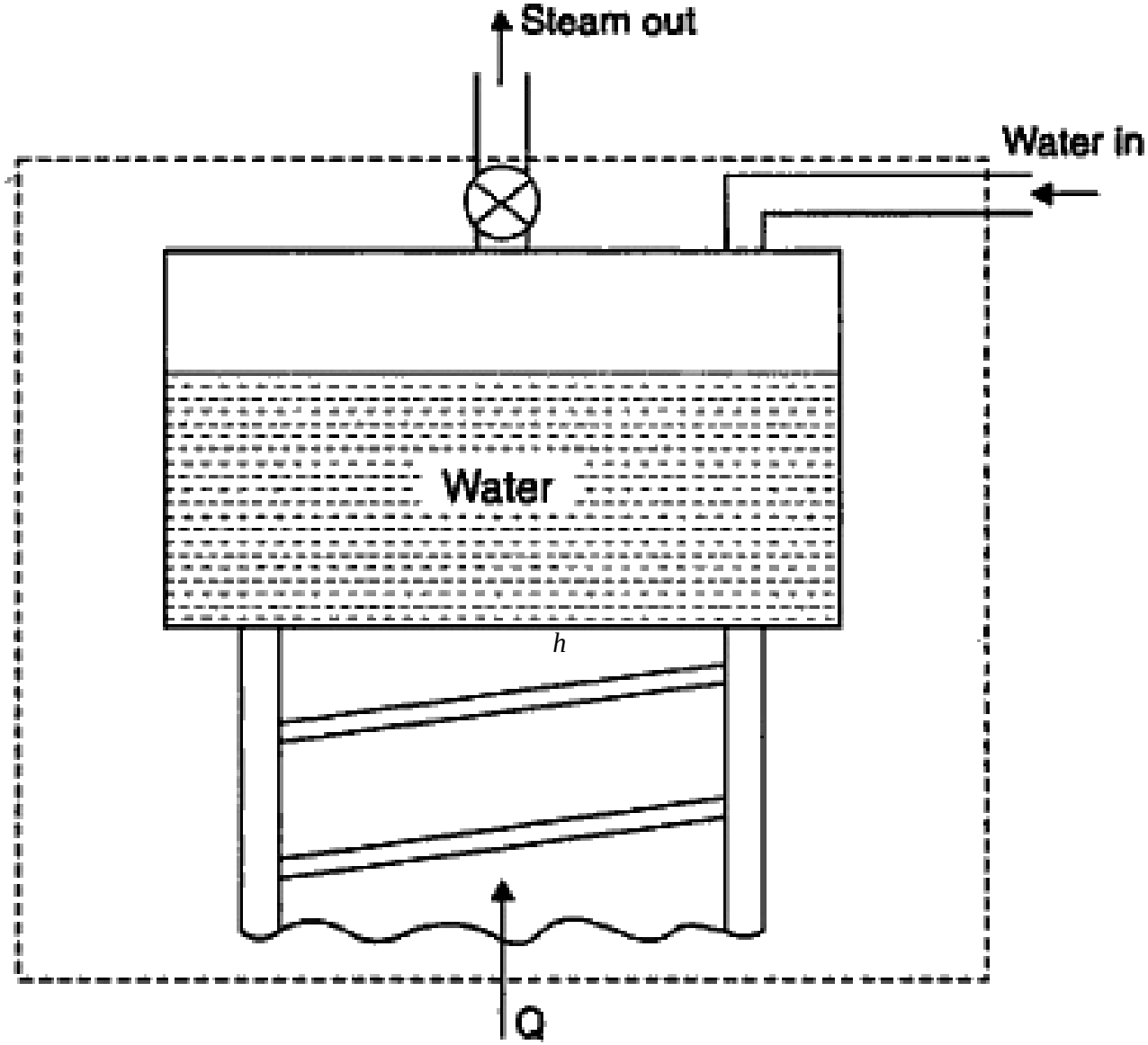
$$W = \dot{m}(h_1 - h_2)$$

Throttling device



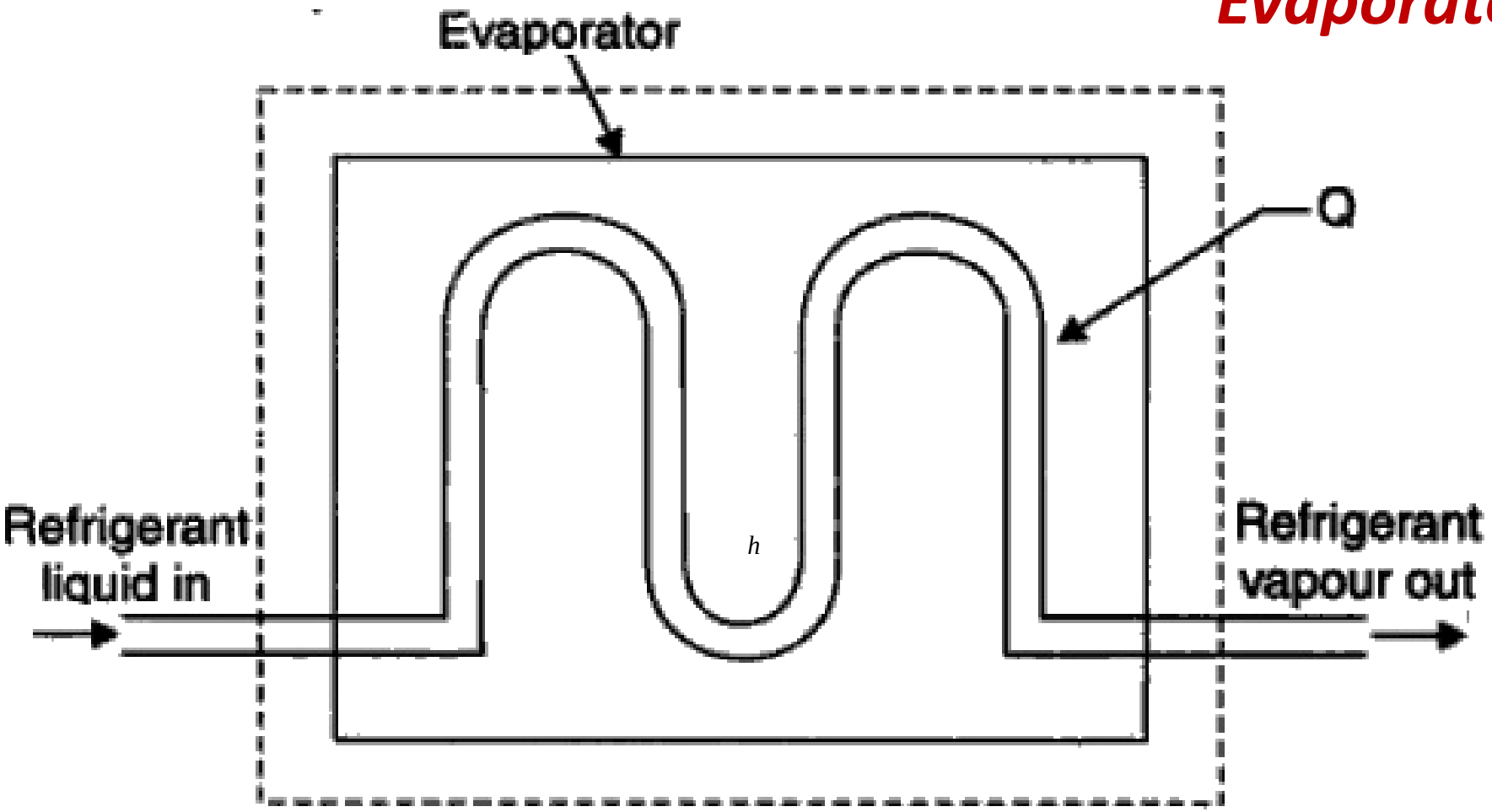
$$h_1 = h_2$$

Boiler



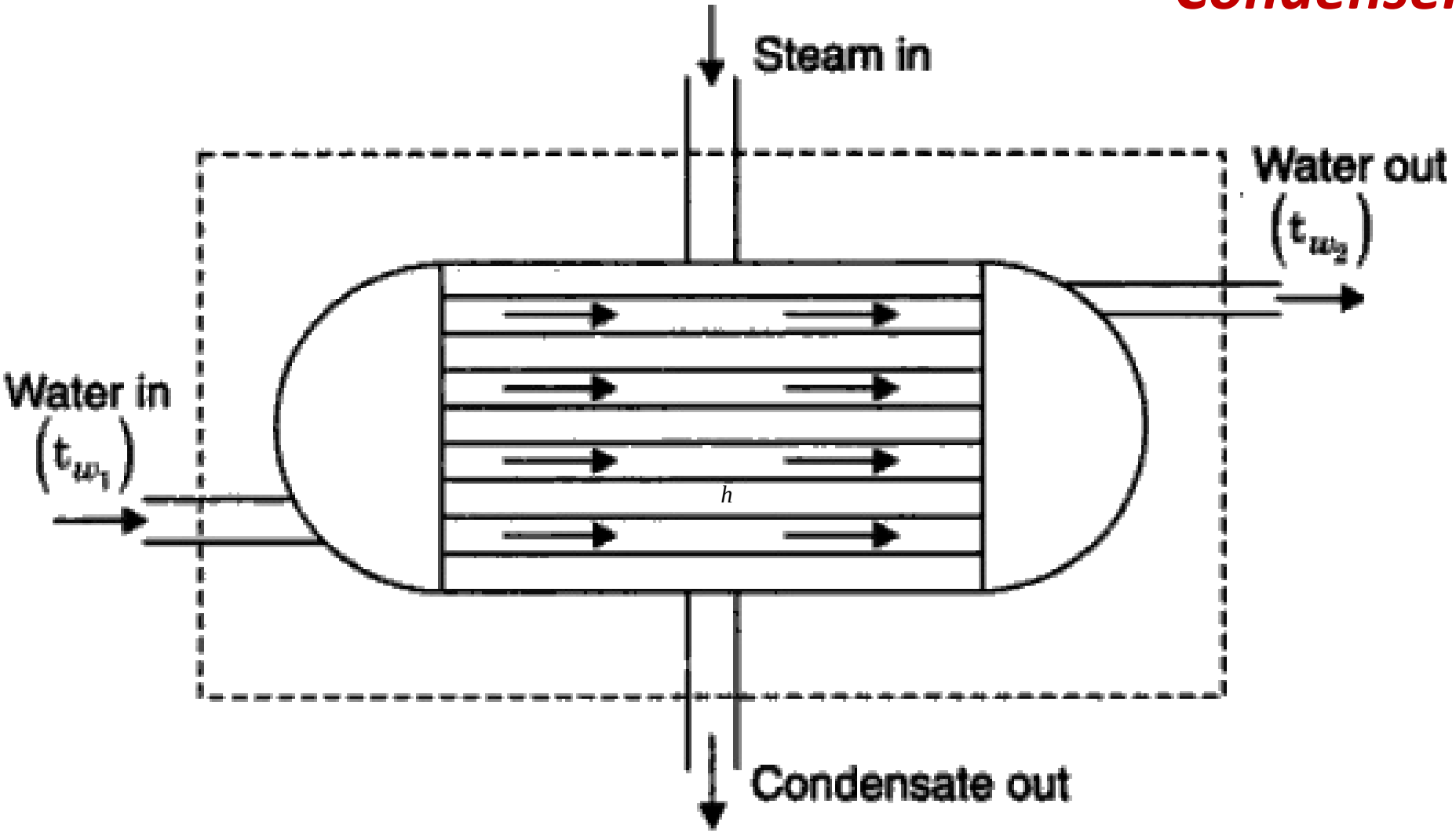
$$Q = \left(h_2 - h_1\right)$$

Evaporator



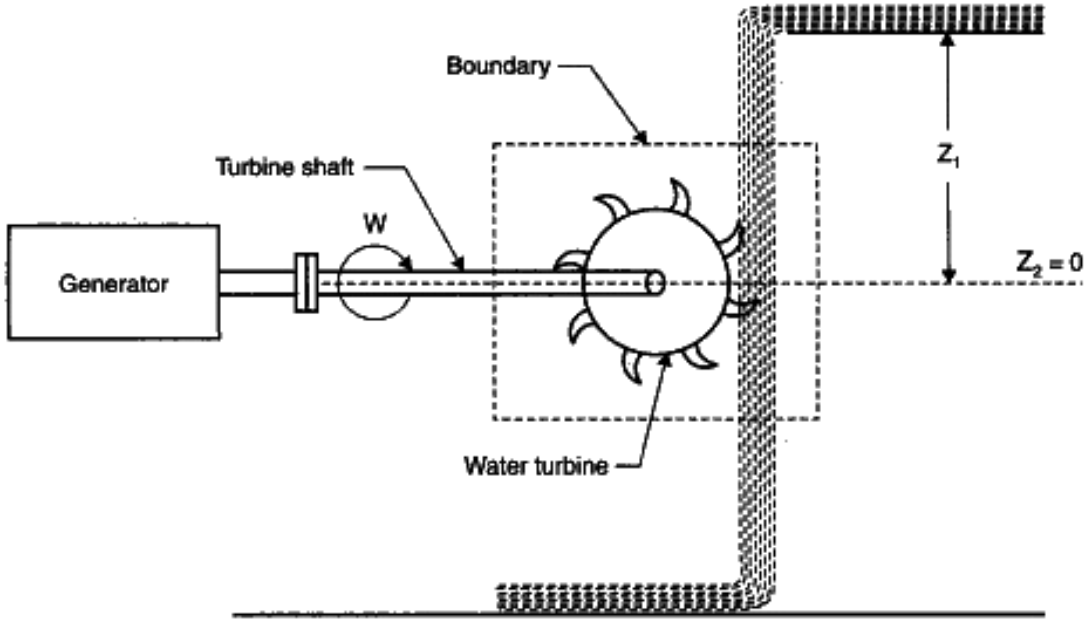
$$Q = \left(h_2 - h_1\right)$$

Condenser



$$Q = (h_1 - h_2)$$

Water Turbine

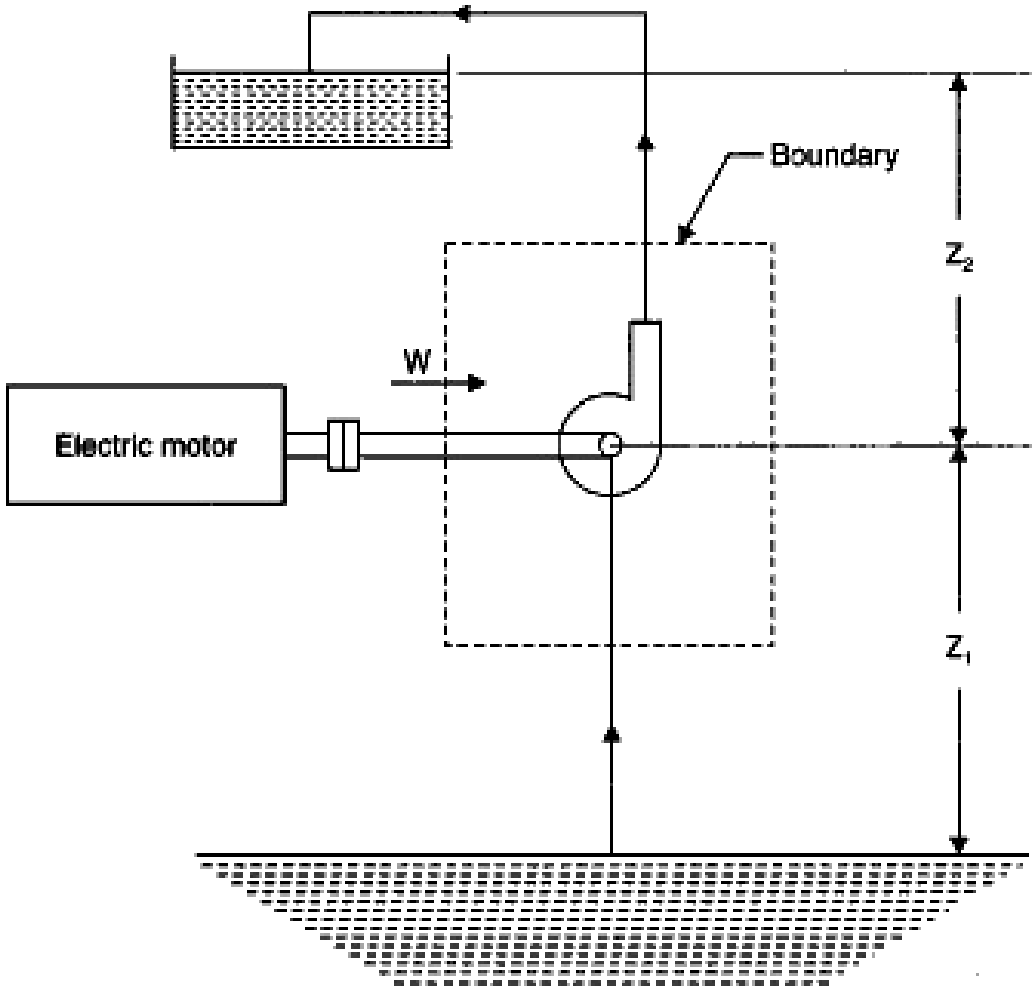


$$\left(u_1 + p_1 v_1 + Z_1 g + \frac{C_1^{\,2}}{2}\right) + Q = \left(u_2 + p_2 v_2 + Z_2 g + \frac{C_2^{\,2}}{2}\right) + W$$

since $Q = 0, \Delta u = 0, v_2 = v_1 = v \text{ and } Z_2 = 0$

$$\boxed{\left(p_1 v + Z_1 g + \frac{C_1^{\,2}}{2}\right) = \left(p_2 v + Z_2 g + \frac{C_2^{\,2}}{2}\right) + W}$$

Water Pump



$$\left(p_1 v_1 + Z_1 g + \frac{C_1^{\,2}}{2}\right) = \left(p_2 v_2 + Z_2 g + \frac{C_2^{\,2}}{2}\right) - W$$

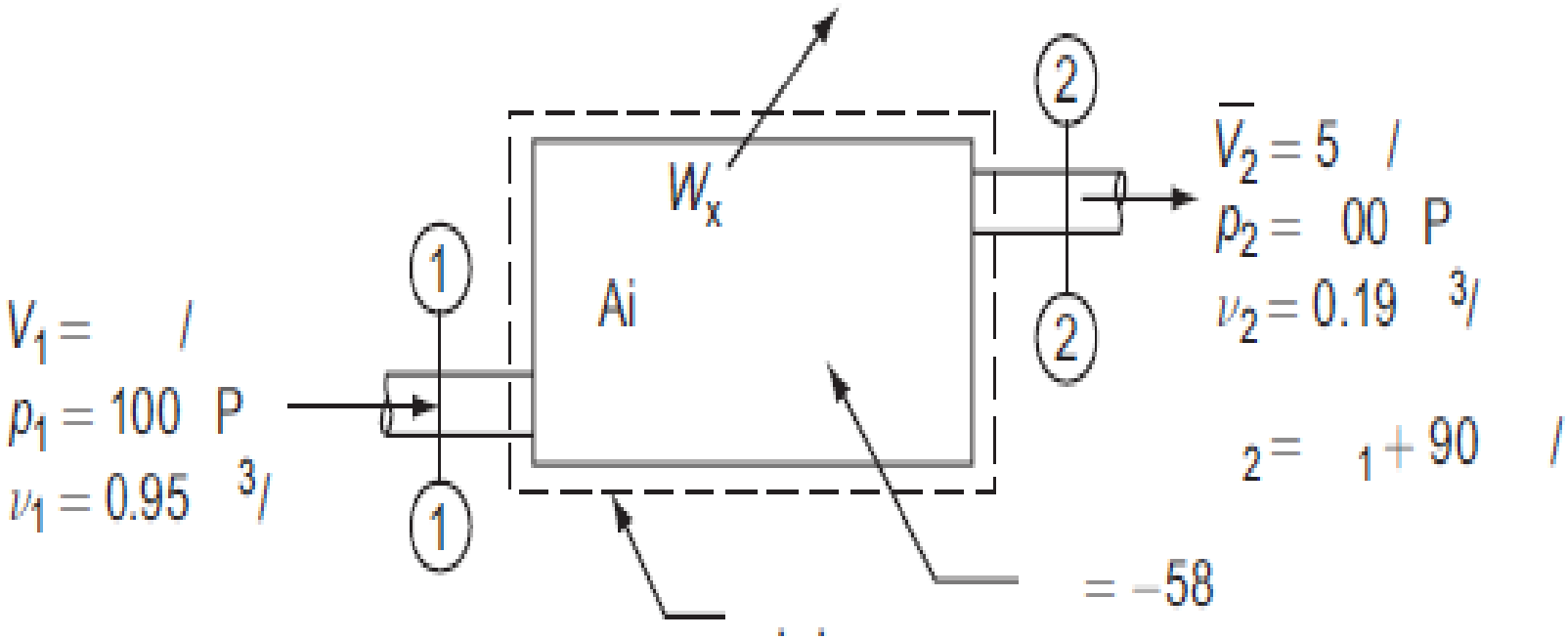
Problem - 1

Air flows steadily at the rate of **0.5 kg/s** through an air compressor, entering at **7 m/s** velocity, **100 kPa** pressure, and **0.95 m^3/kg** volume, and leaving at **5 m/s**, **700 kPa**, and **0.19 m^3/kg**. The internal energy of the air leaving is **90 kJ/kg** greater than that of the air entering. Cooling water in the compressor jackets absorbs heat from the air at the rate of **58 kW**.

(a) Compute the rate of shaft work input to the air in kW.

(b) Find the ratio of the inlet pipe diameter to outlet pipe diameter.

Problem - 1



Problem - 1

(a) Writing the steady flow energy equation, we have

$$w\left(u_1 + p_1\nu_1 + \frac{V_1^2}{2} + {}_1g\right) + \frac{d\,Q}{d\tau} = w\left(u_2 + p_2\nu_2 + \frac{V_2^2}{2} + {}_2g\right) + \frac{d\,W_x}{d\tau}$$

$$\therefore \qquad \frac{d\,W_x}{d\tau} = -w\left((u_2 - u_1) + (p_2\nu_2 - p_1\nu_1) + \frac{V_2^2 - V_1^2}{2} + ({}_2 - {}_1)g\right) + \frac{d\,Q}{d\tau}$$

$$\therefore \qquad \frac{d\,W_x}{d\tau} = -0.5\,\frac{\text{kg}}{\text{s}}\left[90\,\frac{\text{kJ}}{\text{kg}} + (7\times 0.19 - 1\times 0.95)\,100\,\frac{\text{kJ}}{\text{kg}}\right.$$

$$\left. + \frac{(5^2 - 7^2)\times 10^{-3}}{2}\,\frac{\text{kJ}}{\text{kg}} + 0\right] - 58\ \text{kW}$$

$$= -0.5\ \ 90 + 38 - 0.012\ \ \text{kJ/s} - 58\ \text{kW}$$

$$= -122\ \text{kW} \qquad \textit{Ans.}\ (a)$$

Rate of work input is 122 kW.

Problem - 1

(b) From mass balance, we have

$$w = \frac{A_1 \mathbf{V}_1}{\nu_1} = \frac{A_2 \mathbf{V}_2}{\nu_2}$$

$$\therefore \qquad \frac{A_1}{A_2} = \frac{\nu_1}{\nu_2} \cdot \frac{\mathbf{V}_2}{\mathbf{V}_1} = \frac{0.95}{0.19} \times \frac{5}{7} = 3.57$$

$$\therefore \qquad \frac{d_1}{d_2} = \sqrt{3.57} = 1.89 \qquad \textit{Ans.}\ (b)$$

Problem - 2

Air at a temperature of **15 °C** passes through a heat exchanger at a velocity of **30 m/s** where its temperature is raised to **800 °C**. It then enters a turbine with the same velocity of **30 m/s** and expands until the temperature falls to **650 °C**. After leaving the turbine, the air is taken at a velocity of **60 m/s** to a nozzle where it expands until the temperature has fallen to **500 °C**. If the air flow rate is **2 kg/s**, calculate **(a)** the rate of heat transfer to the air in the heat exchanger, **(b)** the power output from the turbine assuming no heat loss, and **(c)** the velocity at exit from the nozzle, assuming no heat loss. Take the enthalpy of air as $\mathbf{h = c_p t}$, where

c_p is the specific heat equal to **1.005 kJ/kg K** and t is the temperature.